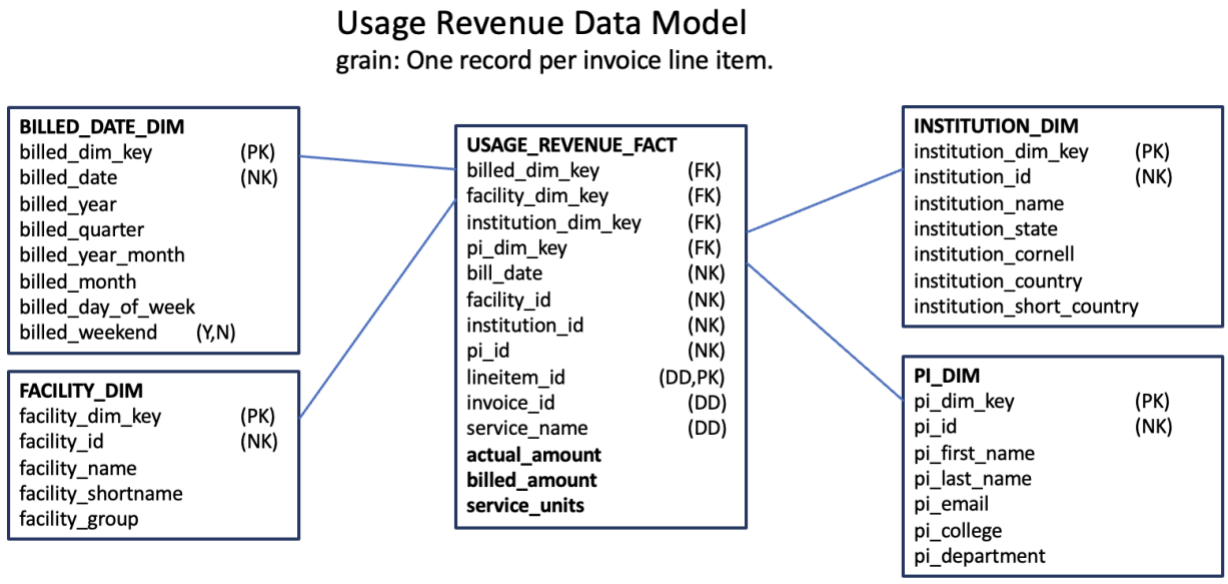
Background:

This WhereScape assignment is based on a prior class project. In this assignment, you will be implementing a dimensional data model and populating the dimensional model from a source system (LIMS) which is a normalized data structure. This assignment will guide you through the process of building need needed dimension and fact table for this dimensional data model. Similar to the WhereScape tutorial assignment, you will see a pattern of working with load table to pull data into your database account. Stage tables for changing column names, column orders, merging data from multiple tables, and adding transformations.



This WhereScape exercise will walk you through the building of each dimension table and finally the fact table. This exercise is meant to prepare you for the class project.

Assuming you have already completed the WhereScape tutorial assignment already, you have a WhereScape catalog in your netid database account.

The grading of this assignment will be based on the completion of your work.
You should address the following questions:

- Are the needed tables in place?
- Do the table and column names match the instructions?
- Are the tables populated?
- Are the data transformations working correctly?

With all of this said, check your work. The exercise does prompt you to "display data" to make sure your loads worked. If there is a transformation, look at the data in that column, does it look correct?

A few tips for this exercise and using WhereScape:

1. If you added the wrong column into load, or missed the step, don't panic, just delete the table and do it again. *To delete an object.. In the left pane (object pane) right click on the table, then select "Delete Metadata and Drop table". You will then be asked if you really want to drop the various components, simply click ok.*
2. Ask yourself: what is the meaning of each process: create(recreate)/update procedure/execute?

   Saying we are creating a dimensional model using WhereScape Red and have already loaded the source data (such as LIMS in the assignment). What's the next step? You need to specify which columns and data types are required by dragging them into the stage table, dimension table, and fact tables. Instead of writing extensive SQL population code yourself, WhereScape Red handles most of the coding automatically. Specifically, each action when you right click during the creation process—such as Create (or Recreate), Update Procedure, and Execute Update Procedure—involves a specific database programming language running in the background. To give you an idea of what WhereScape Red does behind the scenes, we provide some pseudocode for objects like the stage table and dimension table.

   Create(Recreate) process -*WhereScape will execute the SQL to create (or recreate) a table in your database account based on the table definitions you have create in the builder pane.*

```
CREATE TABLE load_sales_transactions (
Column_1 datatype,
Column_2 datatype
);
```

   Update Procedure (Rebuild Update Procedure) -*WhereScape will generated a database stored procedure to populate your table, based on the source-to-target mappings, and transformations, you provided. If more information is needed, such as the join paths for the underlying load_* tables, WhereScape will walk you through a wizard to capture this information. You can view the resulting storted procedures in your database account. Log in with SQL Developer and browse your procedures folder. There will be a stored procedure with naming update_[table_name] for each table you have generated code.*

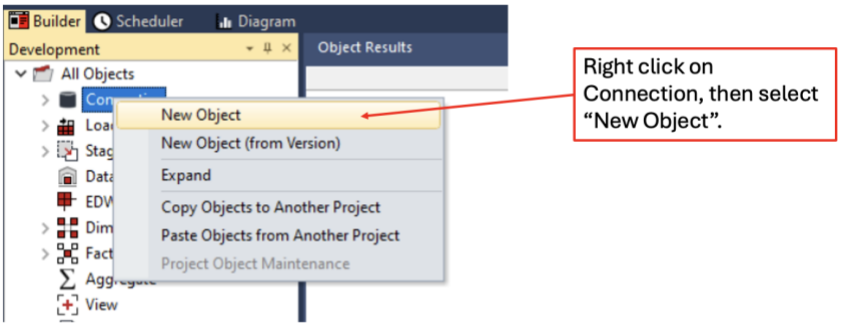
```
CREATE OR REPLACE PROCEDURE procedure_name()
LANGUAGE plpgsql
AS $$
BEGIN
  -- Procedure body
END;
$$;
```

   Execute Update Procedure – *Once you have successfully generated your code, you will need to execute the code to populate your table.*

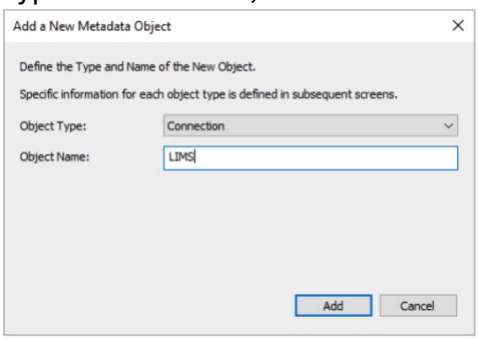
```
CALL procedure_name ();
```

Add new source connection:

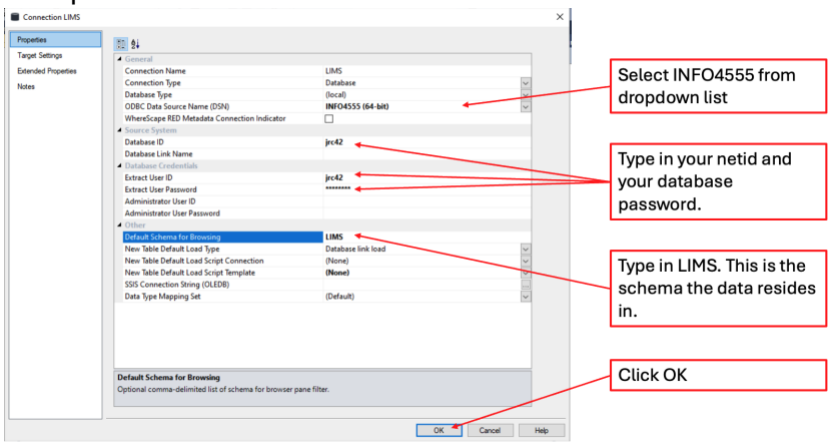
In WhereScape RED, create a new connection for the LIMS (Lab Information Management System) source data.



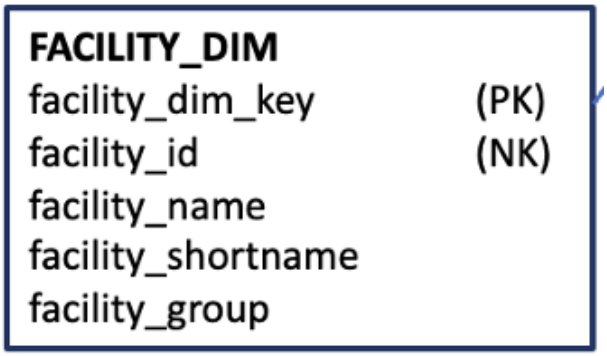
Type name "LIMS", then click Add.



Complete the connection information.



FACILITY_DIM:



Here are the source-to-target mappings for the Facility_dim table:

Source Table	Source Column	Target Table	Target Column	Transformation	Notes
facilities	id	Facility_dim	Facility_id		NK
facilities	name	Facility_dim	Facility_name		
facilities	Short_name	Facility_dim	Facility_shortname		
Facility_groups	Facility_group	Facility_dim	Facility_group		

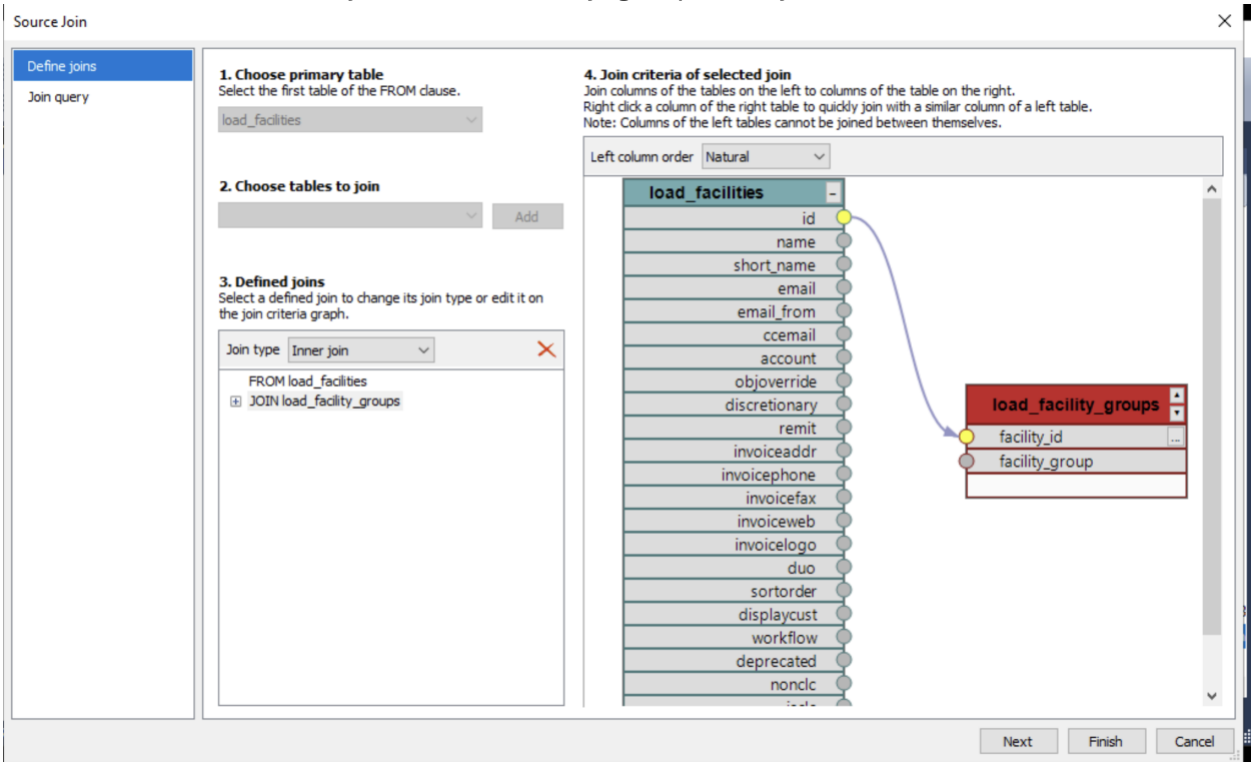
1) Create needed load tables. Looking at the source-to-target mapping table, I have two source tables facilities & facility_groups. Create a load table for each of these tables.
2) Define stage table to join the two load tables.
    a. Double click on Stage tables in left pane, so “stage tables” is the context of center pane.
    b. Change connection in right pane to “Data Warehouse” connection and expand the folder for your netid account.
    c. Expand your load_facilities table and select id and drag to center pane.
    d. You will be prompted for the new table name. I changed the default stage_facilities to simply stage_facility, then click Add.
    e. Click OK and you should then see the “stage_facility columns” screen in the middle.
    f. Drag the following remaining attributes from the load_facilities table, which show up in the above source-to-target mapping, into the center pane:
        i. name
        ii. shortname
    g. Now expand the load_facility_groups table in the right pane and drag the facility_group attribute into the center pane.
    h. In the center pane, double click on the **id** column and rename to **facility_id**.
    i. Rename name to facility_name

j. Rename short_name to facility_shortname
k. Optionally you can right click, and delete, the following columns: dss_create_time & dss_create_date

stage_facility Columns

Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls	Nu...	Ad...
facility_id	facility id	decimal(10)	load_facilities	id	#,##0	Y	Y	Y
facility_name	facility name	nvarchar2(150)	load_facilities	name		Y	N	N
facility_short_name	facility short n...	nvarchar2(50)	load_facilities	short_name		Y	N	N
facility_group	facility group	varchar2(30)	load_facility_groups	facility_group		Y	N	N

3) Create your stage_facility table in your netid database account
   a. Right click on stage_facility, in the left pane and select “Create(ReCreate)”
4) Build code for stage_facility
   a. Right click stage_facility, in left pane, select code -> “rebuild update procedure”
   b. In “Define Stage Procedure Type” screen, select “Set”
   c. In “Select Parameters” screen, select “OK”
   d. In “Source Join” screen, select “Add”, then drag the join arrow from load_facility.id to load_facility_group.facility_id. Should like this:

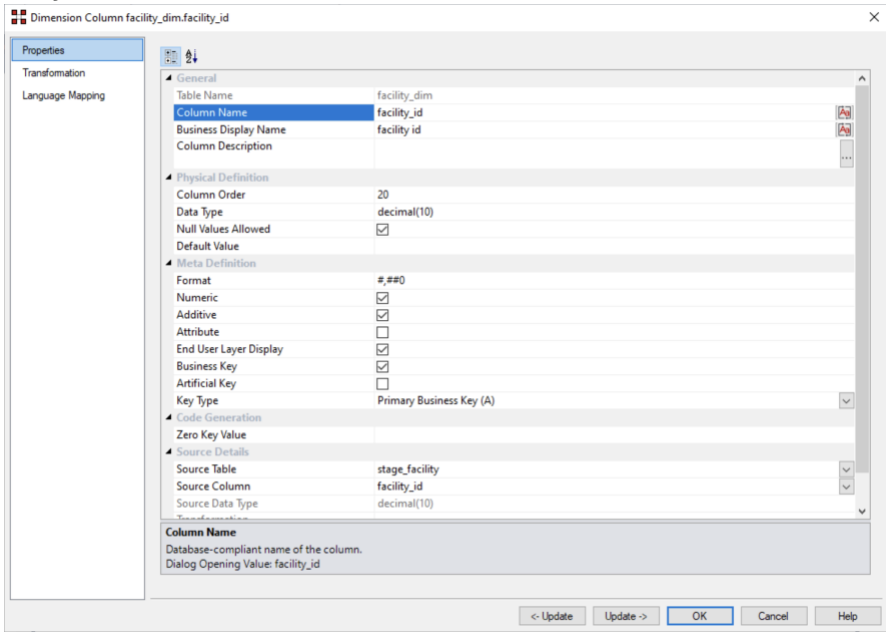


e. Click “Finish”

5) Execute code for stage_facility table. This will load the records for your stage table.

a. Right click stage_facility in the left pane, then select “execute update procedure”. You should see a message in the bottom pane about loading 26 records.
b. Right click stage_facility and select display data, to view the data you just loaded.

6) Define FACILITY_DIM table.
   a. Refresh your warehouse connection in right pane, so your new stage_facility table will show up.
   b. Double click “Dimension” in the left pane, so “Dimension will become the focus of the center pane.
   c. Drag the stage_facility table from the right pane into the center pane
   d. In the “Add a New Metadata Object” screen, rename table from dim_facility to facility_dim to match data model on page 1. Then click Add.
   e. In “dimension type” screen click “Normal”.
   f. In the “dimension facility_dim” screen click OK
   g. In the “Create Database Table” screen click “close”. Not ready to build it yet.
   h. Optionally right click the dss_update_date column and select delete column. I did the same for dss_create_date.
   i. Double click on the facility_id column. Then check the box next to “business key”. You are defining facility_id as the business key (a.k.a natural key) for your dimension.

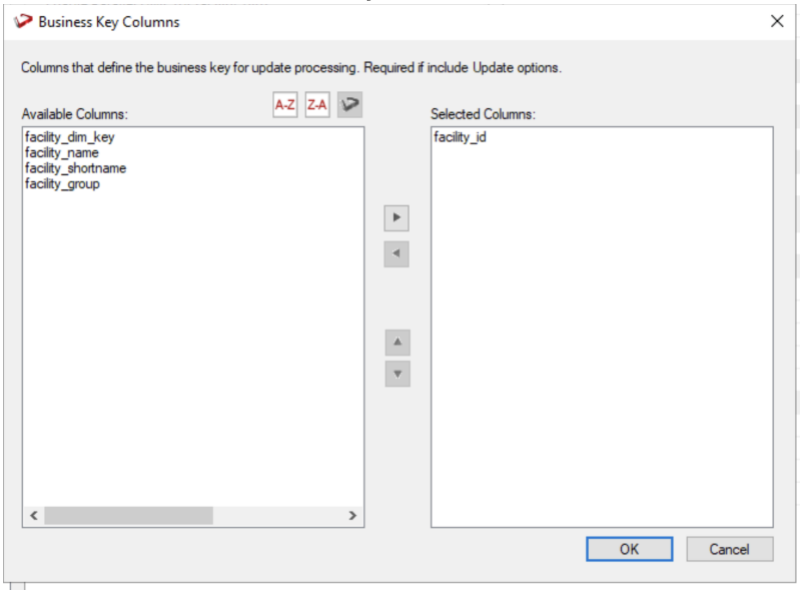


   j. After clicking Business key, click OK
   k. Note that WhereScape created the needed surrogate key for your dimension primary key. Facility_dim_key.

facility_dim Columns

Column Name	Display Name	Data Type	Source Table	Source Column
facility_dim_key	facility dim key	integer		facility_dim_key
facility_id	facility id	decimal(10)	stage_facility	facility_id
facility_name	facility name	nvarchar2(150)	stage_facility	facility_name
facility_short_name	facility short n...	nvarchar2(50)	stage_facility	facility_short_name
facility_group	facility group	varchar2(30)	stage_facility	facility_group

7) Create facility_dim table in your team database account.
    a. Right click on facility_dim, in the left pane and select create(ReCreate)
8) Build code for facility_dim
    a. Right click facility_dim and select rebuild update procedure.
    b. On the “procedure Build Type” screen, the default is SET, click OK
    c. In the “facility_dim update build options” screen, click the button to the right of “Business Key Columns”.
    d. Select facility_id in left side, the click arrow to move to the right side.



    e. Click OK
    f. In facility_dim update build options screen, click OK.
9) Execute the code to load facility_dim
    a. In the left pane, right click on facility_dim, then select “execute update procedure”. You should see a message about loading 26 records.
    b. Right click facility_dim, in right pane, and select display data to see the data you loaded.
10) Right click on your facility_dim and display data. If you see data, you are done.

INSTITUTION_DIM:

INSTITUTION_DIM	
institution_dim_key	(PK)
institution_id	(NK)
institution_name	
institution_state	
institution_cornell	
institution_country	
institution_short_country	

The source-to-target mapping for INSTITUTION_DIM:

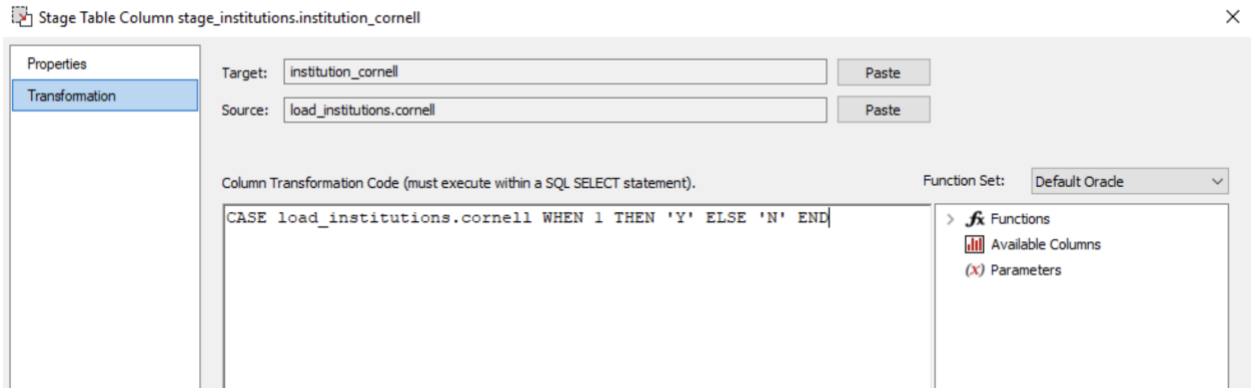
institutions	id	Institution_dim	Institution_id		
institutions	name	Institution_dim	Institution_name		
institutions	state	Institution_dim	Institution_state		
institutions	cornell	Institution_dim	Institution_cornell	If 1 then "Y" else "N"	
countries	name	Institution_dim	Institution_country		
countries	iso	Institution_dim	Institution_short_country		

The institution_dim table has columns from the lims.institutions and lims.countries tables.

1) Create load_tables for institutions and countries
2) Create stage_institutions
    a. Double click stage table folder in left pane, to make stage table the focus of middle pane.
    b. Open warehouse connection in right pane.
    c. Expand your load_institutions table and drag the "id" column into the middle pane.
    d. For the "add new metadata object" I am ok with the name stage_institutions, so click "Add", then click "OK" on next screen.
    e. In the right browsing pane, expand the load_institutions table and drag the columns: name, state, & cornell into the middle pane for the stage_institutions table.
    f. In the middle pane, rename the columns to match the data model
        i. Id -> institution_id
        ii. Name -> institution_name
        iii. State -> institution_state
        iv. Cornell -> institution_cornell
    g. In the right pane, expand your load_countries table and drag the name and iso columns to the middle pane
    h. Rename these new columns to match data model
        i. Name -> institution_country
        ii. Iso -> institution_short_country

i. Change cornell indicator flag from (0,1) to (N,Y) to match data model
   i. Double click the institution_cornell column
   ii. Change the Data Type from decimal(3) to varchar2(1)
   iii. Uncheck the numeric and additive attributes.
   iv. In upper left corner, click the “transformation” tab
   v. Add this case statement to the transformation box:

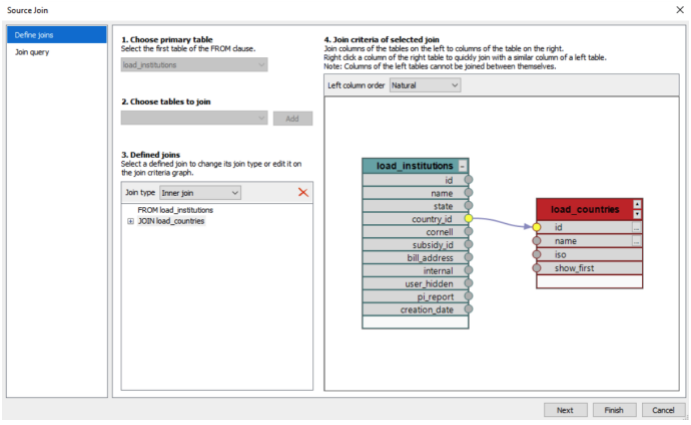
CASE load_institutions.cornell WHEN 1 THEN 'Y' ELSE 'N' END



   vi. Click OK

j. Create stage_institutions table in your account
   i. In left pane, right click stage institutions and select Create(Recreate).

k. Build code for stage_institutions table
   i. In left pane, right click stage_institutions and select code > “Rebuild Update Procedure”
   ii. Select SET
   iii. In “Select parameters” screen, Click OK
   iv. In “source join” screen, select Add
   v. Join load_insitutions.country_id with load_countries.id

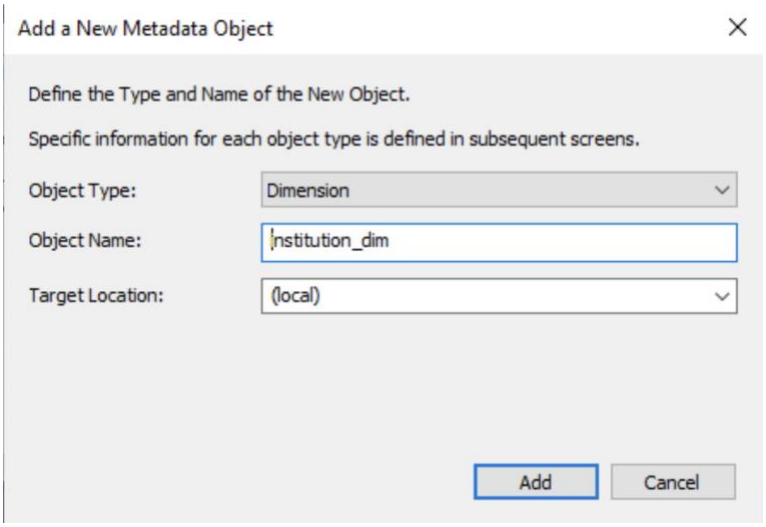


   vi. Click “Finish”

l. Load your stage_institutions table
   i. In left pane, right click stage_insitutions and select “execute update procedure”

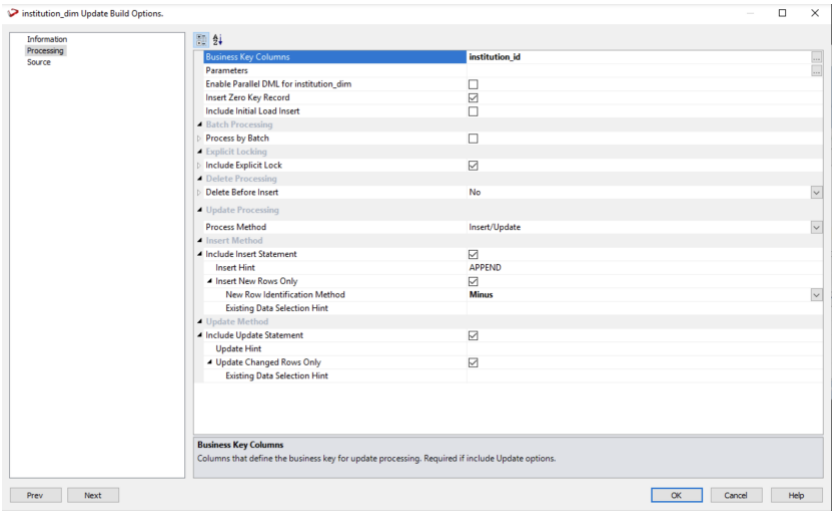
ii. In left pane, right click stage_institutions and select “display data” Does the data look as you would expect? Did your transformation work for Y/N?

3) Create INSTITUTION_DIM table
   a. In left pane, click “Dimension” to make dimension the focus of middle pane
   b. In right pane, refresh your data warehouse connection
   c. In right pane, select stage_institutions and drag to middle pane
   d. In the “Add a New Metadata Object” screen, rename table to match your data model.



   e. Click Add
   f. For dimension type, select “normal”
   g. In the dimension INSTITUTION_DIM screen, simply click OK
   h. In the “Create Database Table” screen, click “Create”.

4) Generate code for INSTITUTION_DIM table
   a. In left pane, right click institution_dim and select code > rebuild update procedure
   b. For procedure build type screen, select OK
   c. Set your business key column (NK) to institution_id, then click OK



5) Load INSTITUTION_DIM table
   a. In left pane, right click institution_dim, then select execute update procedure
   b. Should load 1032 records
   c. In left pane, right click institution_dim, then select “display data”. Does the data look as you would expect?

PI_DIM:

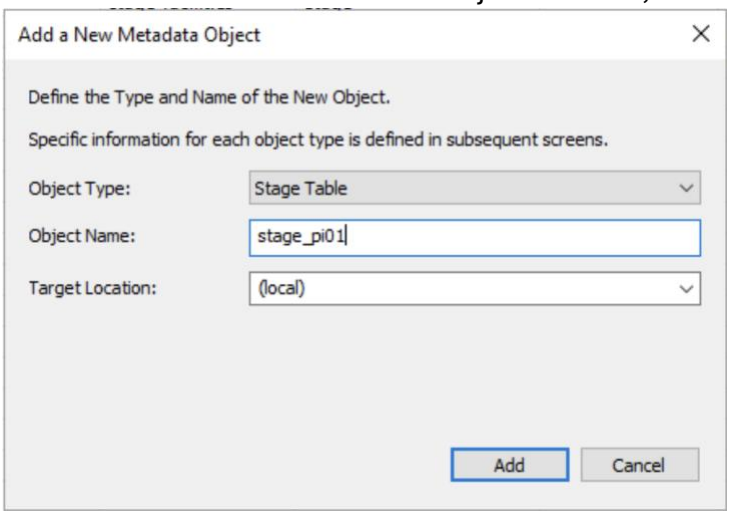
PI_DIM	
pi_dim_key	(PK)
pi_id	(NK)
pi_first_name	
pi_last_name	
pi_email	
pi_college	
pi_department	

The source-to-target mapping for the PI_DIM table:

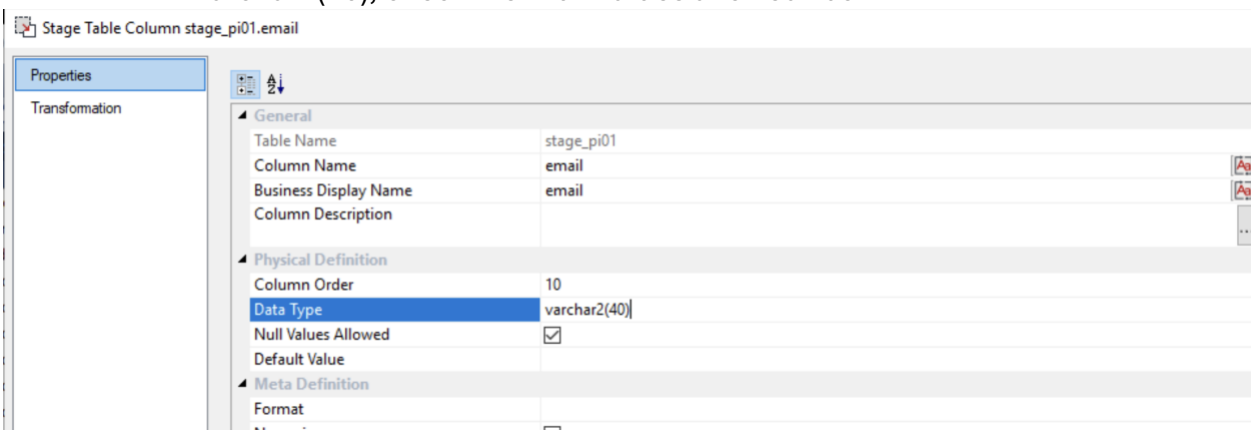
pis	id	Pi_dim	Pi_id		NK
pis	First_name	Pi_dim	Pi_first_name		
pis	Last_name	Pi_dim	Pi_last_name		
pis	email	Pi_dim	Pi_email		
Cu_person	Primary_college_org_name	Pi_dim	Pi_college		
Cu_person	Primary_org_name	Pi_dim	Pi_department		

As you can see from the source-to-target mapping table, there are two source tables, from the lims system, *pis* and *cu_person*. Start by creating load tables for these two tables.

1) Create load_cu_person and load_pis tables
2) Create stage_pi table from load_cu_person table. This will be an opportunity to change names to match model and add email to cu_person data, which will be the business key to join with pis data.
    a. In left pane, click STAGE Table to make STAGE Table the focus of middle pane
    b. In right pane change connection to Data Warehouse.
    c. Select load_cu_person and drag into middle pane.
    d. In the "Add a New Metadata Object" screen, name stage table stage_pi01

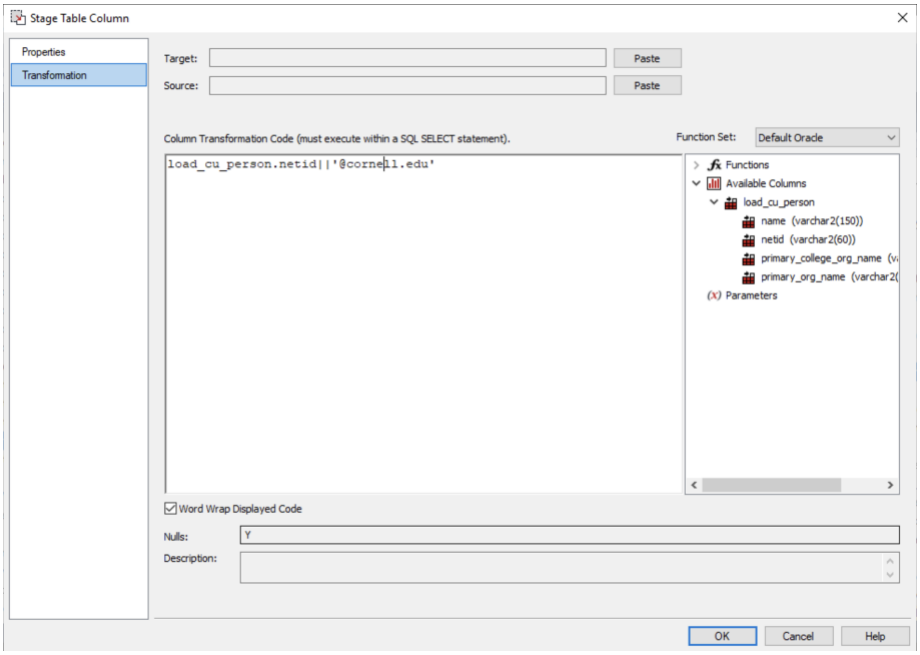


    e. Click Add
    f. Click OK
    g. In the middle pane, right click and select "add column"
    h. For column name and business display name type "email". For data type use varchar2(40), check the "null values allowed" box.



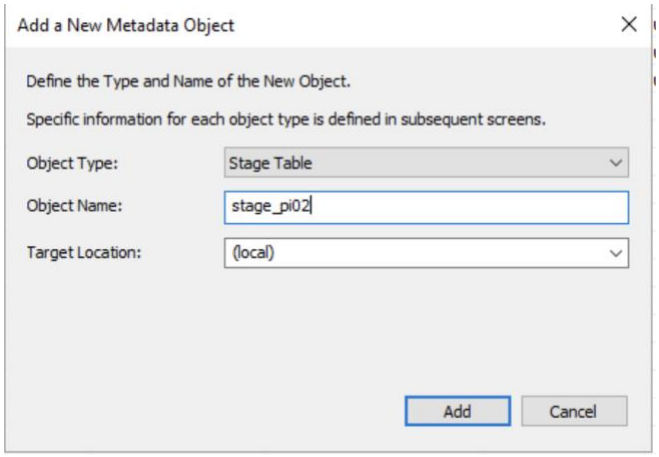
    i. In the upper left side, select "Transformation"
    j. In the transformation box, you will want to enter the following:
    load_cu_person.netid||'@cornell.edu'

This is simply appending @cornell.edu to the netid, to give the cornell email address.



k. After adding the transformation, click OK
l. Now, in the columns screen, rename the following two columns:
    i. Primary_college_org_name -> college
    ii. Primary_org_name -> department
m. Now create the stage_pi01 table you have defined
    i. In the left pane, right click stage_pi01, then select Create(Recreate)
n. Now generate the code
    i. In the left pane, right click stage_pi01, then select code> Rebuild Update Procedure
    ii. Click SET
    iii. In the "select parameters" screen, simply click OK
o. Load your stage_pi01 table
    i. In the left pane, right click stage_pi01 and select "execute update procedure"
    ii. You should see a message about loading 562,681 records.
    iii. In the left pane, right click stage_pi01 and select "display data". How does the data look? Is the email column populated correctly?

3) Create stage_pi02 table
    a. In the left pane click "Stage Table", to make Stage Table the focus of the middle pane
    b. In the right pane, refresh your warehouse connection
    c. drag the load_pis table from the right pane to the center pane
    d. In the "Add a New Metadata Object" screen, rename to "stage_pi02", then click Add



e. Click OK
f. In the "stage_pi02 Columns" screen, delete all of the columns not in the pi_dimension. After deleting columns, should look like:

stage_pi02 Columns

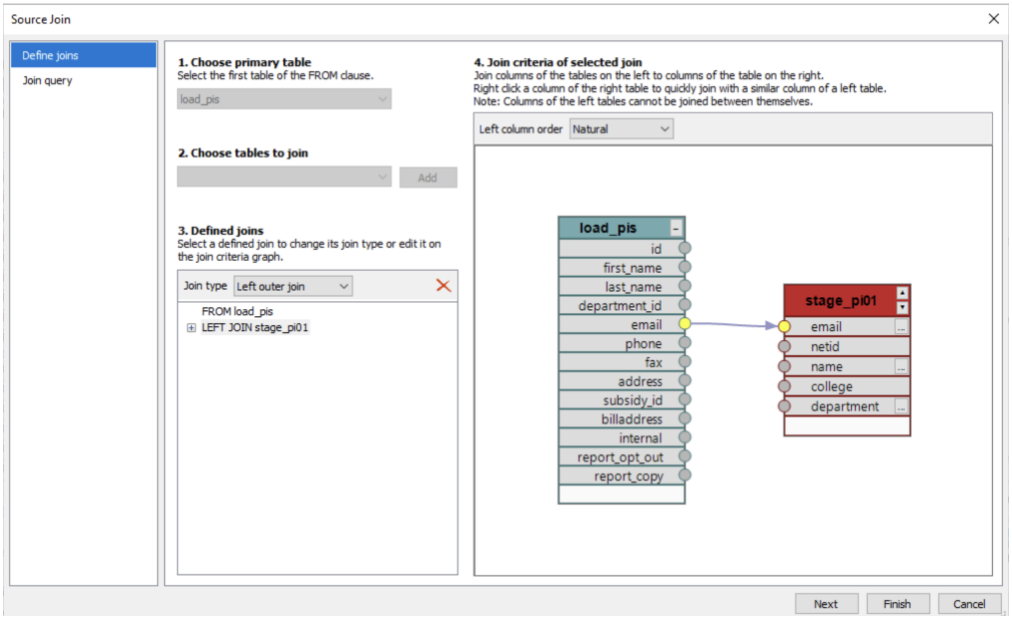
Column Name	Display Name	Data Type	Source Table	Source Column	Format	N
id	id	decimal(10)	load_pis	id	#,##0	Y
first_name	first name	nvarchar2(50)	load_pis	first_name		Y
last_name	last name	nvarchar2(50)	load_pis	last_name		Y
email	email	nvarchar2(75)	load_pis	email		Y

g. In the right pane, expand your stage_pi01 table and drag the college and department columns into the middle pane.
h. Now rename all of the columns to match the columns in the pi_dim table. Basically just add a prefix of "pi_" to each column name.

stage_pi02 Columns

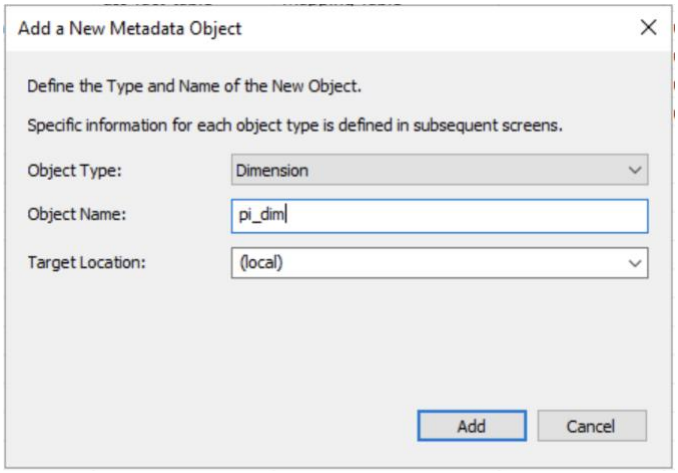
Column Name	Display Name	Data Type	Source Table	Source Column	Format	N
pi_id	pi id	decimal(10)	load_pis	id	#,##0	Y
pi_first_name	pi first name	nvarchar2(50)	load_pis	first_name		Y
pi_last_name	pi last name	nvarchar2(50)	load_pis	last_name		Y
pi_email	pi email	nvarchar2(75)	load_pis	email		Y
pi_college	pi college	varchar2(120)	stage_pi01	college		Y
pi_department	pi department	varchar2(120)	stage_pi01	department		Y

i. Create stage_pi02 table in database
    i. In left pane, right click stage_pi02 and select Create(recreate)
j. Create code
    i. In left pane, right click stage_pi02 and select code>"rebuild update procedure"
    ii. Select SET
    iii. In "select parameters" screen select OK
    iv. In "source join" screen, click Add to define the join
    v. Join the two tables on email, select "left outer join", then click Finish

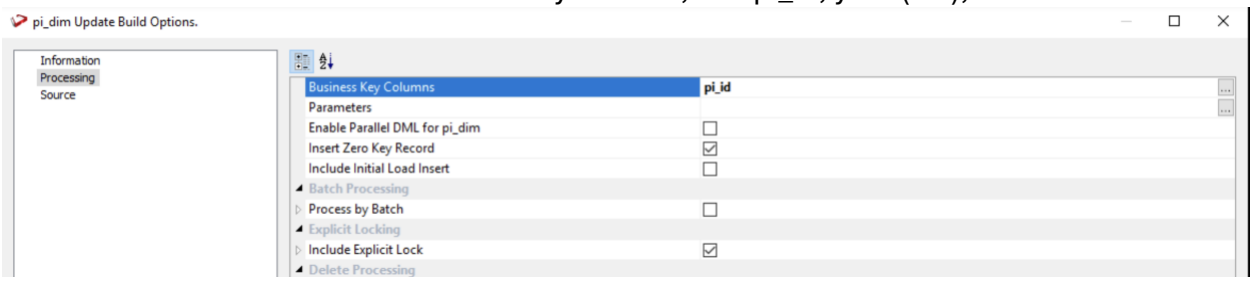


k. Load stage_pi02 table
   i. In left pane, right click stage_pi02 table and select “execute update procedure”
   ii. You should now have 3,942 records in your stage_pi02 table, the same record count as in your load_pis table
   iii. In left pane, right click stage_pi02 table and select “display data”. How does the data look. You will notice most of the records do not have a college or department value, since they are not Cornell Pis. However, you should be able to find PI records with college and department values.

4) Create PI_DIM table
   a. In left pane, click “dimension” to make dimension the focus of the middle pane
   b. In right pane, refresh your data warehouse connect, so your new stage_pi02 table will show up.
   c. Drag stage_pi02 from the right pane to the middle pane.
   d. In “Add a New Metadata Object” screen, rename object name to match data model. Then click Add



e. For dimension type, select normal.
f. Select OK
g. Select “Create”
h. Create code
    i. In left pane, pi_dim, then select code>rebuild update procedure
    ii. For build type, select OK
    iii. For the business key column, use pi_id, your (NK), then click OK



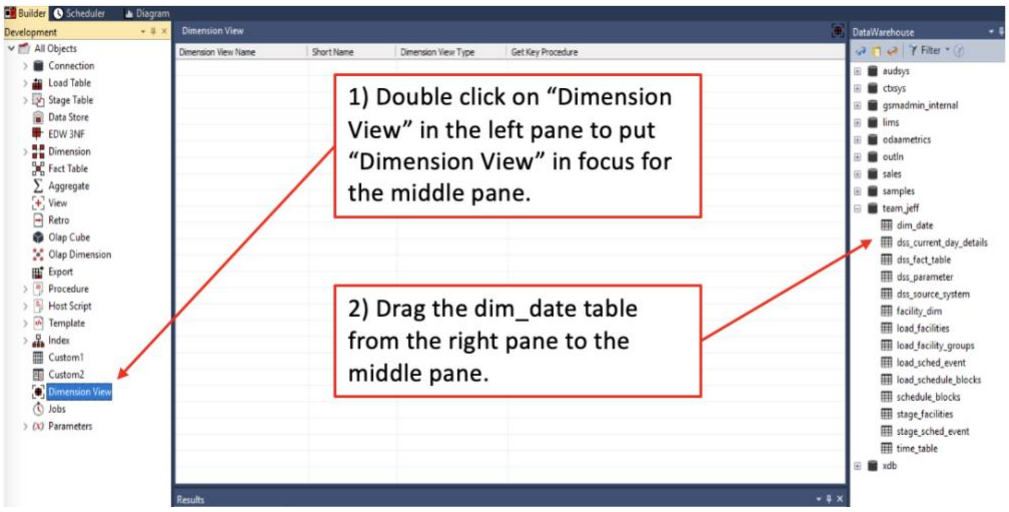
i. Load pi_dim table
    i. In the left pane, right click and select “execute update procedure”
    ii. The load process should load 3,942 records, matching your stage_pi02 table.
    iii. In left pane, right click pi_dim and select “display data”. How does the data in your pi_dimension look?

BILLED_DATE_DIM:

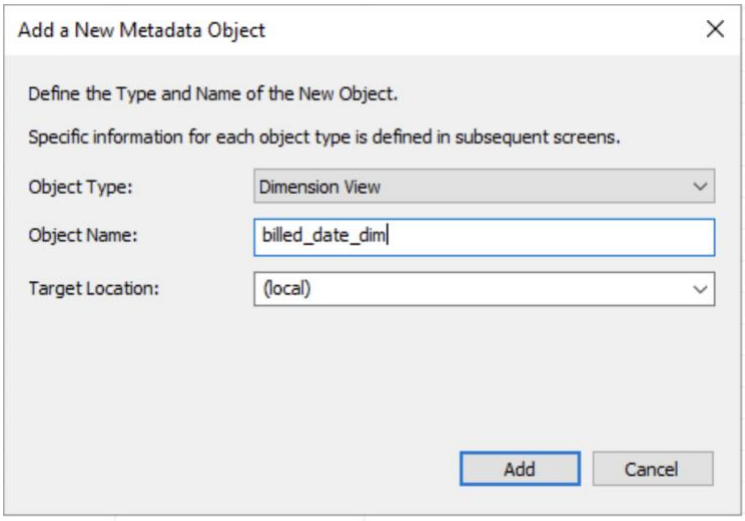
BILLED_DATE_DIM	
billed_dim_key	(PK)
billed_date	(NK)
billed_year	
billed_quarter	
billed_year_month	
billed_month	
billed_day_of_week	
billed_weekend	(Y,N)

Unlike the other dimensions in this data model, the billed_date_dim is a role play view built over the delivered WhereScape dim_date table. This section walks you through the steps to create a custom role play view, over the dim_date table. Be careful to not update, or drop, the delivered WhereScape dim_date table. You are creating a dimension view, not replacing the dim_date table.

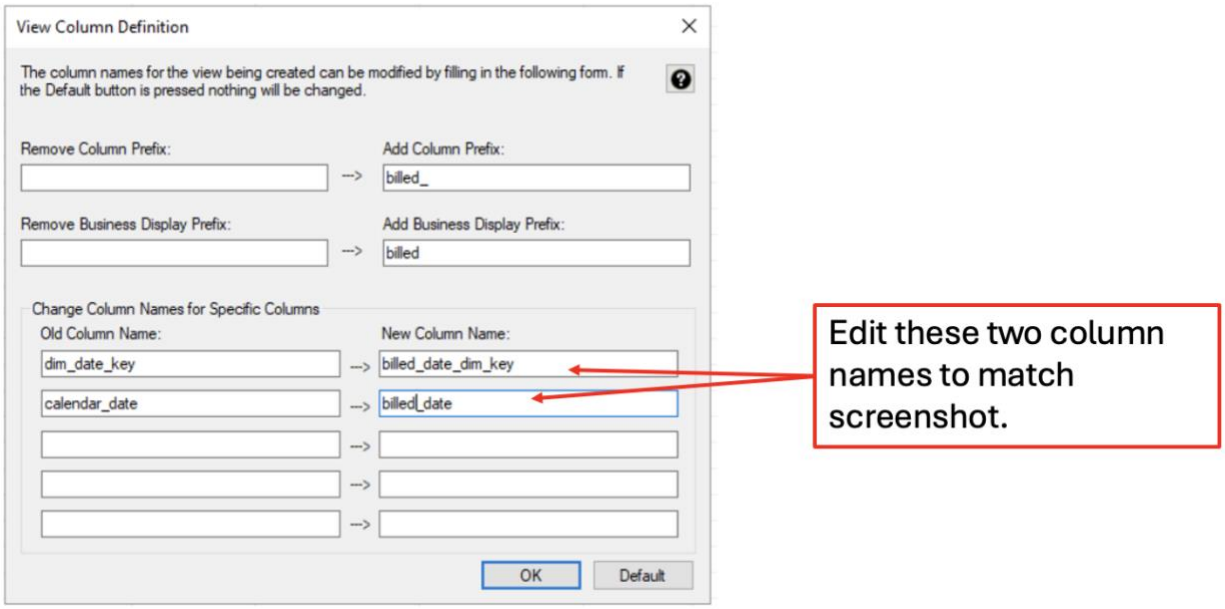
Start by double clicking on “dimension view” in the left pane to put “dimension view” in focus for the middle pane. Next expand your connection, under your warehouse connection, and drag your dim_date table into the middle pane. This indicates to WhereScape that you want to build a dimension view over the dim_date table.



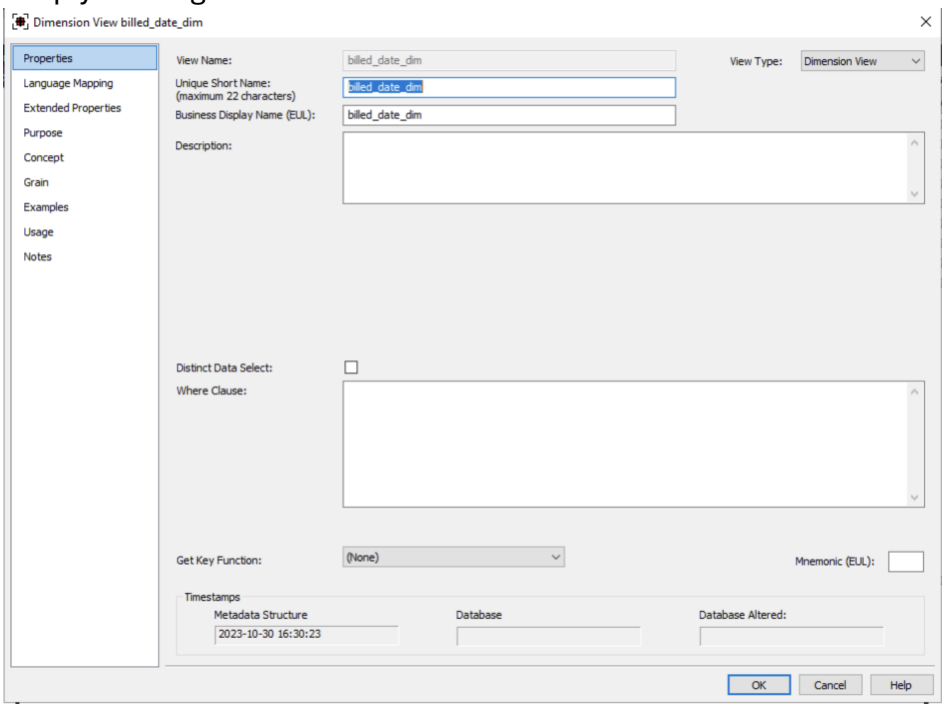
In the “Add a New Metadata Object” panel, rename the object name to match your dimension name, then click Add.



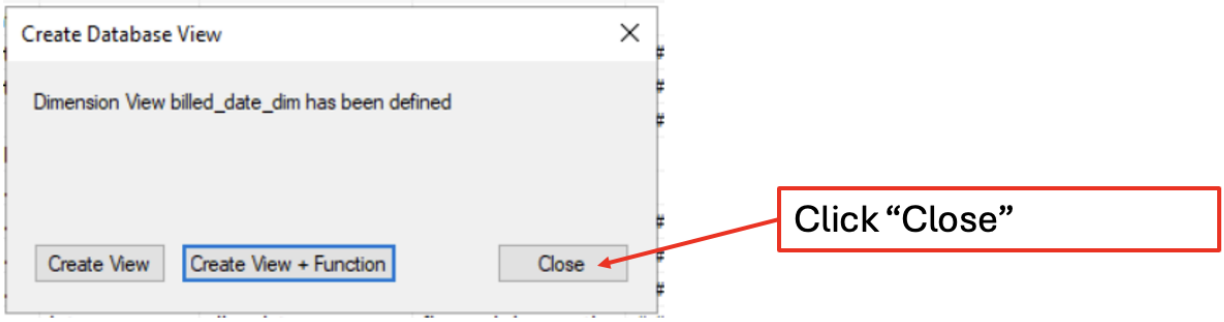
In the “view Column Definition” panel, does the proposed prefix match your model? Note that the default mapping for calendar_date is calendar_date. Rename the new column to billed_date to match the data model. Also, I renamed the key to “billed_date_dim_key”. Then click OK.



For the Dimension View panel, you could put a filter in the where clause window, but I'm simply clicking OK.



For the "create Database View" panel, simply click Close. We still have some editing to do, before creating the view.



Note there are many columns in the delivered dim_date table which are not helpful given our requirements. Honestly, I'm not sure what some of them even mean. So, with some reviewing of the dim_date table, via SQL Developer, I found the following attribute mapping, between my billed_date_dim and the delivered dim_date table. All other attributes I will drop from my dimension view.

Source_table	Source_column	Target_table	Target_column

Dim_date	Dim_date_key	Billed_date_dim	Billed_date_dim_key
Dim_date	Calendar_date	Billed_date_dim	Billed_date
Dim_date	Cal_year	Billed_date_dim	Billed_year
Dim_date	Cal_quarter	Billed_date_dim	Billed_quarter
Dim_date	Cal_month	Billed_date_dim	Billed_yearmonth
Dim_date	Cal_month_name	Billed_date_dim	Billed_month
Dim_date	Cal_day_in_week	Billed_date_dim	Billed_day_of_week
Dim_date	Cal_day_in_week_no	Billed_date_dim	Billed_day_of_week_no
Dim_date	Week_end_flag	Billed_date_dim	Billed_week_end_flg

In the middle pane, select and drop any column, which does not exist in the source column of the above source-to-target mapping. Note you should determine the source columns in dim_date for your date dimension table.

After deleting the many columns I don't care about, it looked like this.

billed_date_dim Columns

Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls	Nu...	Ad...
billed_date_dim_key	billed date dim...	integer	dim_date	dim_date_key	#,##0	N	Y	Y
billed_date	billed date	date	dim_date	calendar_date		Y	N	N
billed_year	billed year	integer	dim_date	cal_year	#,##0	Y	Y	Y
billed_quarter	billed quarter	integer	dim_date	cal_quarter	#,##0	Y	Y	Y
billed_yearmonth	billed yearmon...	integer	dim_date	cal_month	#,##0	Y	Y	Y
billed_month	billed month	varchar2(7)	dim_date	cal_month_name		Y	N	N
billed_cal_day_of_week	billed cal day o...	varchar2(3)	dim_date	cal_day_in_week		Y	N	N
billed_cal_day_of_week_no	billed cal day o...	integer	dim_date	cal_day_in_week_no	#,##0	Y	Y	Y
billed_week_end_flg	billed week en...	varchar2(1)	dim_date	week_end_flag		Y	N	N

Note that the column names still don't match my data model. I don't have "_cal_" in my data model column names. All of my dates are calendar dates versus financial, etc, so I simply want billed_year, billed_quarter, etc. To change the name of a column, double click the column name, then you may edit it. Also, there are a few columns in this screenshot I missed in my deletes, so I deleted them afterwards. The column order didn't match my data model either, so I rearranged them by dragging and dropping them up or down.

To create your new dimension view in your team database account, right click on billed_dim_date, in the left pane, then select Create(Replace).

Now your billed_dim_date will show up in your netid account as a database view. However, there is one more step needed, before you are done. Right click on billed_date_dim, in the left pane, and select code-> "Rebuild Get Key Procedure". This step is needed for dimension views to work properly with your fact table.

On the build option screen, simply click OK.

Check out your newly built date dimension view by right clicking billed_date_dim, in the left pane, then selecting display data.

USAGE_REVENUE_FACT:

USAGE_REVENUE_FACT	
billed_dim_key	(FK)
facility_dim_key	(FK)
institution_dim_key	(FK)
pi_dim_key	(FK)
bill_date	(NK)
facility_id	(NK)
institution_id	(NK)
pi_id	(NK)
lineitem_id	(DD,PK)
invoice_id	(DD)
service_name	(DD)
actual_amount	
billed_amount	
service_units	

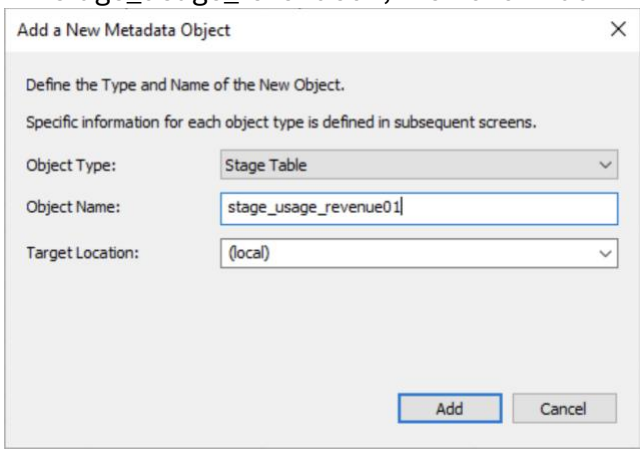
Here's the source-to-target mappings for the USAGE_REVENUE_FACT table:

Source Table	Source Column	Target Table	Target Column	Transformation Logic	Notes
invoice	billdate	Usage_revenue_fact	Billed_date		NK
invoice	Pi_id	Usage_revenue_fact	Pi_id		NK
invoice	Institution_id	Usage_revenue_fact	Institution_id		NK
lineitem	Facility_id	Usage_revenue_fact	Facility_id		NK
lineitem	id	Usage_revenue_fact	Lineitem_id		DD
lineitem	Invoice_id	Usage_revenue_fact	Invoice_id		DD
lineitem	name	Usage_revenue_fact	Service_name		DD
lineitem	Line_total_raw	Usage_revenue_fact	Actual_amount		
lineitem	Line_total	Usage_revenue_fact	Billed_amount		
lineitem	quantity	Usage_revenue_fact	Service_units		

1) Create load tables for each needed source table

a. From the above source-to-target mapping table, there are only two source tables: invoice and lineitem
b. Create load tables for invoice and lineitem

2) Create STAGE_USAGE_REVENUE01 table
   a. In left pane, click on “Stage Table”, so stage table becomes the focus of the middle pane.
   b. In right pane, change your connection to Data Warehouse.
   c. In the right pane, expand your load_invoice table, select billdate and drag it to the middle pane. Note that I only pulled one column over, because I don’t want the many columns from invoice, which add no value to my fact table.
   d. In the “Add a New Metadata Object” screen, rename the object name to stage_usage_revenue01, then click Add

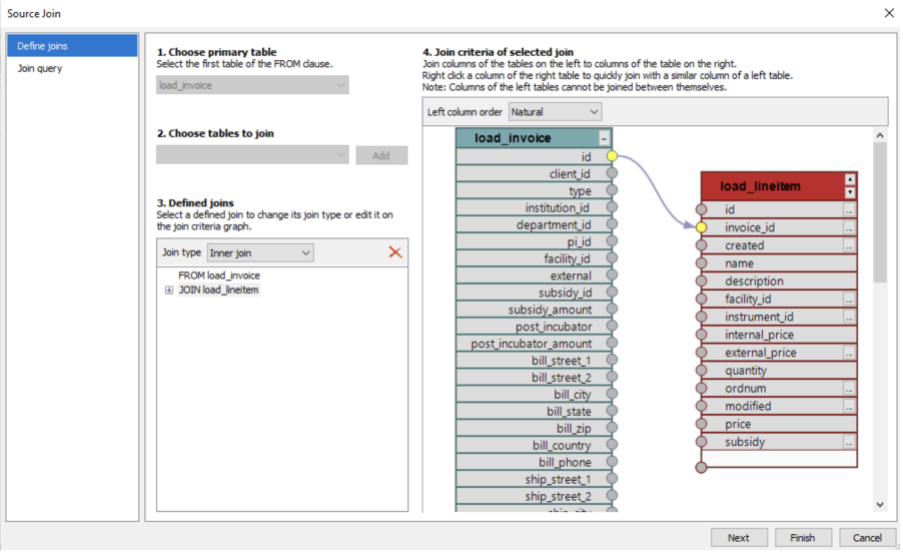


   e. Click OK
   f. From the right pane, drag pi_id from the load_invoice table to the middle pane. Pi_id is another (NK) needed, since there is a relationship to the pi_dim table.
   g. From the right pane, drag institution_id from the load_invoice table to the middle pane. This is another (NK) from the invoice table.
   h. From the right pane, expand your load_lineitem table and drag the following columns to the middle pane. Note, these are the remaining source columns from your source-to-target mapping.
      i. Id
      ii. Invoice_id
      iii. Name
      iv. Facility_id
      v. Quantity
      vi. Line_total_raw
      vii. Line_total
   i. Optionally delete the two audit columns dss_create_time and dss_update_time from your table.
   j. In the middle pane, rename the following columns, so they match the data model. Note these are the mappings from the source-to-target document.

i. Billdate -> billed_date
ii. Id -> lineitem_id
iii. Name -> service_name
iv. Quantity -> service_units
v. Line_item_raw -> actual_amount
vi. Line_total -> billed_amount

Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls	Nu...	Ad...	Att...
billed_date	billed date	date	load_invoice	billdate		Y	N	N	Y
pi_id	pi id	decimal(10)	load_invoice	pi_id	#,##0	Y	Y	Y	N
institution_id	institution id	decimal(10)	load_invoice	institution_id	#,##0	Y	Y	Y	N
lineitem_id	lineitem_id	decimal(10)	load_lineitem	id	#,##0	Y	Y	Y	N
invoice_id	invoice id	decimal(10)	load_lineitem	invoice_id	#,##0	Y	Y	Y	N
service_name	service name	nvarchar2(150)	load_lineitem	name		Y	N	N	Y
facility_id	facility id	decimal(10)	load_lineitem	facility_id	#,##0	Y	Y	Y	N
service_units	service units	decimal(20,2)	load_lineitem	quantity	#,##0	Y	Y	Y	N
actual_amount	actual amount	decimal(20,2)	load_lineitem	line_total_raw	#,##0	Y	Y	Y	N
billed_amount	billed amount	decimal(20,2)	load_lineitem	line_total	#,##0	Y	Y	Y	N

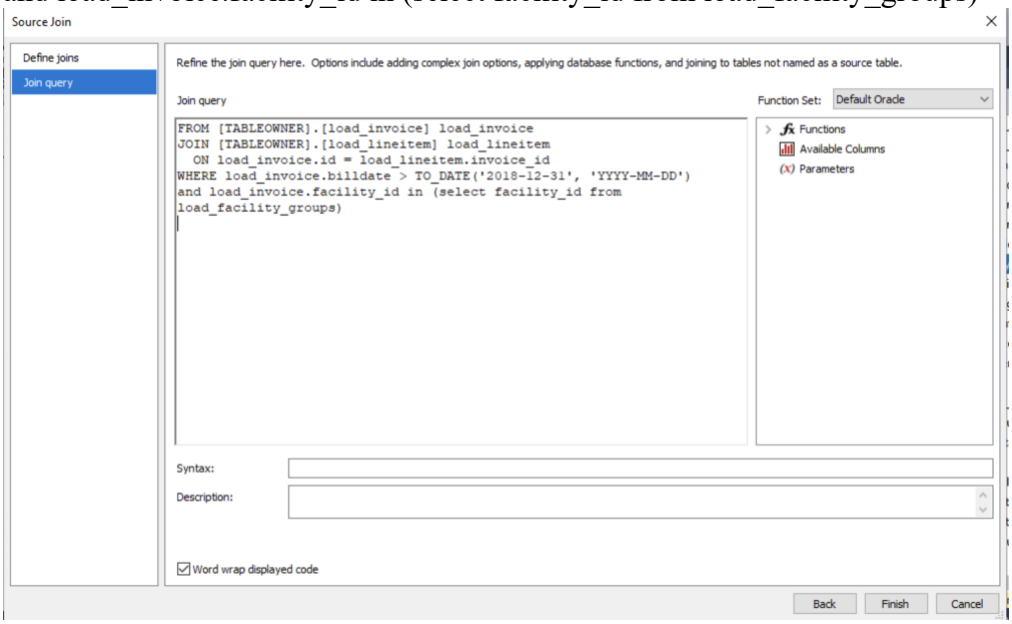
k. Create stage_usage_revenue01 table in database
   i. In left pane, right click on stage_usage_revenue01 and select Create(Recreate)
l. Create code for stage_usage_revenue01 table
   i. In left pane, right click on stage_usage_revenue01 and select code > “rebuild update procedure”
   ii. In the “Define Stage Procedure Type” screen, check box for “Allow Modification of the Where Clause”. For procedure type, click SET
   iii. For “select parameters”, simply click OK
   iv. In “source join” screen, click Add
   v. Join load_invoice.id to load_lineitem.invoice_id, then click Next. *See screenshot below.*



vi. Add the following filter to end of the join query code. This will filter the data in your fact table to records after December 2018. Five full years.

The second part of the filter is limiting the facility_ids to those mentioned in the project document. They live in the facility_groups table.

WHERE load_invoice.billdate > TO_DATE('2018-12-31', 'YYYY-MM-DD')
and load_invoice.facility_id in (select facility_id from load_facility_groups)



vii. Click Finish

m. Load your stage_usage_revenue01 table.
   i. In left pane, right click on stage_usage_revenue01 and select “execute update procedure”
   ii. You should see a message about loading 107,154 records.
   iii. In left pane, right click on stage_usage_revenue01 and select “display data”. How does the data look?

n. Add the surrogate keys (primary keys) from each of the four dimensions.
   i. In left pane, right click on stage_usage_revenue01, select “display columns”
   ii. In the right pane, expand billed_date_dim, select billed_date_dim_key and drag it to the middle pane.
   iii. In the right pane, expand facility_dim, select facility_dim_key and drag it to the middle pane.
   iv. In the right pane, expand institutions_dim, select institutions_dim_key and drag it to the middle pane.
   v. In the right pane, expand pi_dim, select pi_dim_key and drag it to the middle pane.
   vi. Now organize the order of the columns, by drag and dropping them, in the middle pane. Generally, fact tables are ordered with the FKs and the top, followed by the NKs, then DDs, and finally the facts. After I moved my columns to match the data model, it looked like this.

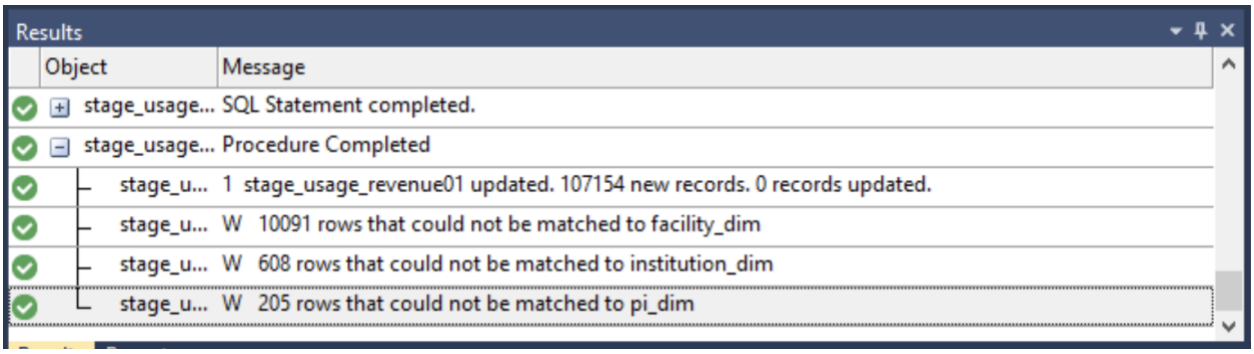
stage_usage_revenue01 Columns

Column Name	Display Name	Data Type	Source Table	Source Column	Format	Nulls	Nu...	Ad...	Att...	Ke...	Order
billed_date_dim_key	billed date dim...	integer	billed_date_dim	billed_date_dim_key	#,##0	N	Y	Y	N		10
facility_dim_key	facility dim key	integer	facility_dim	facility_dim_key	#,##0	N	Y	N	N		20
institutions_dim_key	institutions di...	integer	institutions_dim	institutions_dim_key	#,##0	N	Y	N	N		30
pi_dim_key	pi dim key	integer	pi_dim	pi_dim_key	#,##0	N	Y	N	N		40
billed_date	billed date	date	load_invoice	billdate		Y	N	N	Y		50
facility_id	facility id	decimal(10)	load_lineitem	facility_id	#,##0	Y	Y	Y	N		60
institution_id	institution id	decimal(10)	load_invoice	institution_id	#,##0	Y	Y	Y	N		70
pi_id	pi id	decimal(10)	load_invoice	pi_id	#,##0	Y	Y	Y	N		80
lineitem_id	lineitem_id	decimal(10)	load_lineitem	id	#,##0	Y	Y	Y	N		90
invoice_id	invoice id	decimal(10)	load_lineitem	invoice_id	#,##0	Y	Y	Y	N		100
service_name	service name	nvarchar2(150)	load_lineitem	name		Y	N	N	Y		110
actual_amount	actual amount	decimal(20,2)	load_lineitem	line_total_raw	#,##0	Y	Y	Y	N		120
billed_amount	billed amount	decimal(20,2)	load_lineitem	line_total	#,##0	Y	Y	Y	N		130
service_units	service units	decimal(20,2)	load_lineitem	quantity	#,##0	Y	Y	Y	N		140

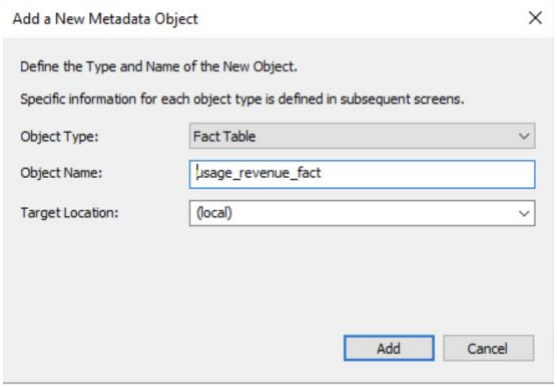
o. Recreate the stage_usage_revenue01 table in database
   i. Now that we have added the surrogate keys (FKs) and re-arranged the column order, we need to synch our new table metadata with the table in the database.
   ii. In left pane, right click stage_usage_revenue01 and select “Create(Recreate)”
   iii. For the Create(Recreate screen, simply click OK

p. Rebuild the code for the stage_usage_revenue01 table.
   i. In left pane, right click on stage_usage_revenue01 table then select code > “rebuild update_stage_usage_revenue01”
   ii. Click SET
   iii. Click OK
   iv. Assuming the initial load of your table worked, you do not need to change the join, or filter, definition in the join screen, so simply click Finish
   v. Next you will be prompted for the business key mappings (NKs) for each of the four dimensions. If the business key column names match, between the stage table and the dimensions, WhereScape will auto populate these business keys. If the business key is populated, go ahead and click OK for each. If any of the joins are missing the business key, find the appropriate column in the list.

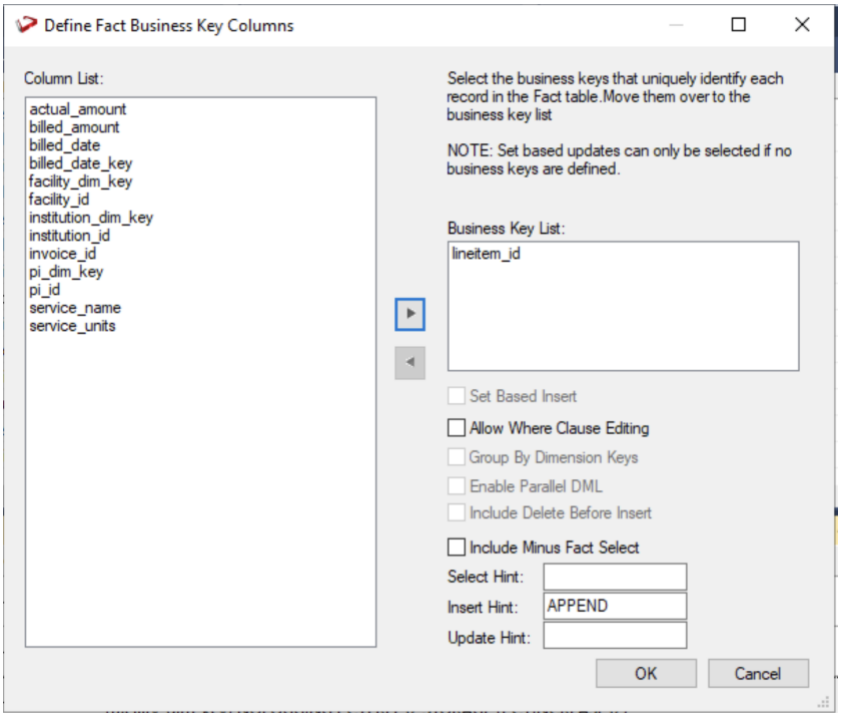
q. Load the stage_usage_revenue01 table
   i. In the left pane, right click stage_usage_revenue01 table and select “execute update procedure”
   ii. In the results pane, you should see a message similar to below. The records that could not be matched will be the fact records which join to a dimension 0 key value.



3) Create USAGE_REVENUE_FACT table
    a. In left pane, click on “Fact Table” to make Fact Table the focus of middle pane.
    b. In right pane, refresh the Data Warehouse connection
    c. In the right pane, select the stage_usage_revenue01 table and drag it into the middle pane
    d. In the “Add a New Metadata Object” screen, rename to match data model. Click



    e. Click Add
    f. Click OK
    g. Click Create
    h. Create code
        i. In left pane, right click on usage_revenue_fact and select code > “rebuild update procedure”
        ii. Define the business key for the fact table. Lineitem_id is a unique identifier.



iii. Click OK

i. Load usage_revenue_fact table

i. In left pane, right click usage_revenue_fact and select "execute update procedure" **Note this will run a minute or two.**

ii. In the results pane you should see a message about loading 107,154 records.

4) Does your fact table now contain data? Are your surrogate key columns correctly populated?

The grading of this assignment will be based on the completion of your work.
You should address the following questions:

- **Are the needed tables in place?**
- **Do the table and column names match the instructions?**
- **Are the tables populated?**
- **Are the data transformations working correctly?**

With all of this said, check your work. The exercise does prompt you to "display data" to make sure your loads worked. If there is a transformation, look at the data in that column, does it look correct?

Turning in your work. To indicate you have completed the assignment, create a schema diagram of your usage_revenue_fact table and upload the image to Canvas.

